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EE1030/EE2100: Matrix Theory

Lecture 17: K-Means Clustering

Topics :

1. Grouping Subproblem
 2. Identification of Representative vectors
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Idea of clustering

given: Collection of Vectors $\{\vec{x}_1, \dots, \vec{x}_m\} \in \mathbb{R}^n$

↓

group into 'k' clusters (k is given)

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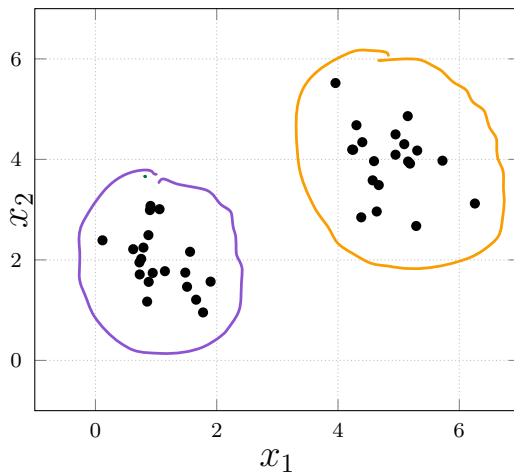
assign to each vector $\leftarrow$ a label $c \rightarrow$ group that it belongs to

Vectors in a group are close to the one's in the group & far from the vectors in the other group

(by distance b/w vectors)

$\hookrightarrow || \quad || \leftarrow L_2 \text{ norm.}$

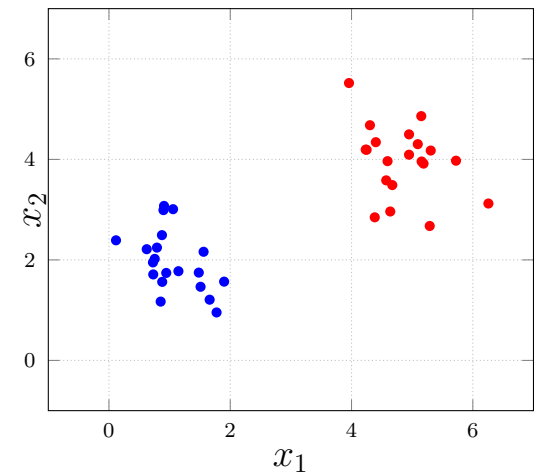
Example: $n=2$



(a) Unlabeled

Looking $\vec{x}_1, \dots, \vec{x}_m$
as points in 2D plane

clustering seems
trivial



(b) Labeled

Fig. 1: A two-dimensional data set with two clusters.

Grouping Subproblem

Start with: given a vector, which group should it belong to?

case a) if each group has only one vector is it?

$$c = \arg \min_i \|\vec{x}_{\text{new}} - \vec{x}_i\|$$

only vector is the group

Case b) if the groups have more than 1 vector?

assume each group can be effectively rep by

a vector $\vec{r}_i \rightarrow$ representative vector of group 'i'

$$c = \arg \min_i \|\vec{x}_{\text{new}} - \vec{r}_i\|$$

for every vector 'i' in set

$$c_i = \arg \min_{j \in (1, \dots, K)} \|\vec{x}_{\text{new}} - \vec{r}_j\|$$

representative vector $\rightarrow$ a) delivery hub
(may or may not)

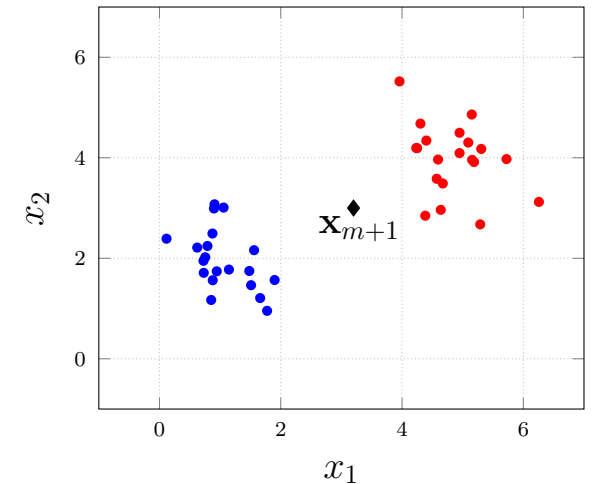


Fig. 2: Which cluster does the new point $\mathbf{x}_{m+1}$ belong to?

Identification of Representative Vectors

Assumption: Vectors in the group are known

goal: to compute $\vec{r}_i$

group $i = \{\vec{x}_{i,1}, \dots, \vec{x}_{i,n_i}\} \leftarrow \text{no. of vectors} = n_i$

one approach: $\vec{r}_i$ is that vector that minimizes the sum of ^{or squared distance} distance to all elements in the group.

$$\vec{r}_i : \min \sum_{i=1}^{n_i} \|\vec{x}_i - \vec{r}_i\|^2 = E \rightarrow \textcircled{1}$$

analytical solution = $\vec{r}_i$ is that vector $\frac{dE}{d\vec{r}_i} = \vec{0}$

$$\hookrightarrow \text{gradient}^T = \begin{bmatrix} \frac{dE}{dr_{i,1}} \\ \vdots \\ \frac{dE}{dr_{i,n}} \end{bmatrix} = \vec{0}$$

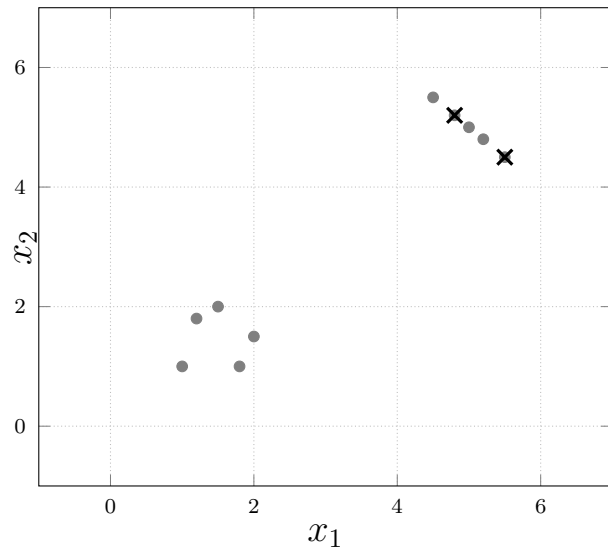
Solve for $\textcircled{1}$ by employing the grad condition

$$\vec{r}_i = \frac{1}{n_i} \sum_{i=1}^{n_i} \vec{x}_i$$

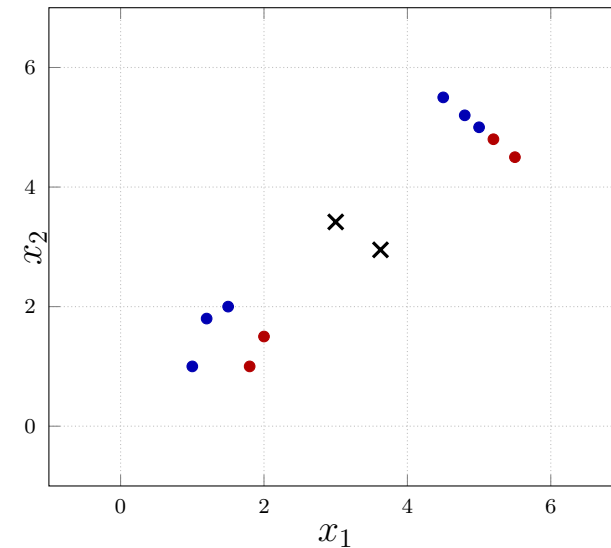
K-Means Clustering Algorithm

- ① Step 1: Start with random $\vec{r}_i \leftarrow$ (free to pick first $\vec{x}_1, \dots, \vec{x}_k$ of the data)
- ② group the data points
- ③ for the groups from step ②, compute $\vec{r}_i$ for each group
- ④ the convergence, repeat steps ② & ③
 $\downarrow$
 grouping does not change $\Rightarrow \vec{r}_i$ does not change $\forall i = 1, \dots, m$
 $\downarrow$
 $(\|\vec{r}_i\| < \epsilon) \forall i$

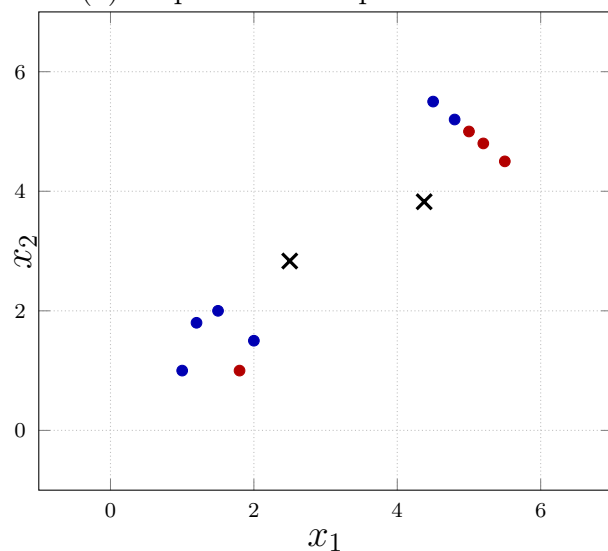
Illustration of Self-Correcting Nature of K-Means Clustering Algorithm



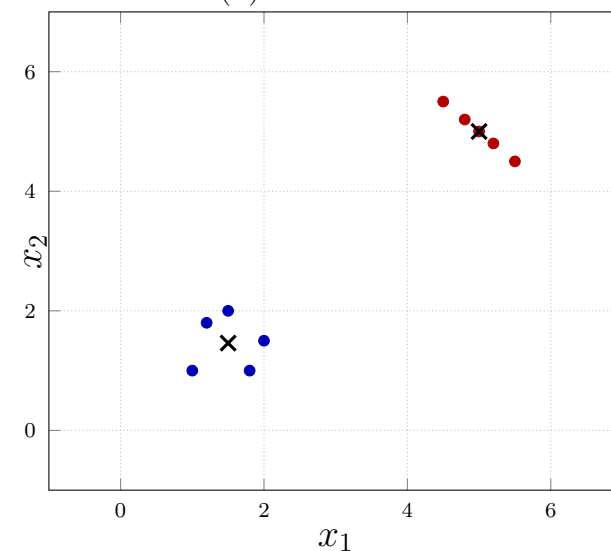
(a) Step 0: Initial representatives



(b) Iteration 1



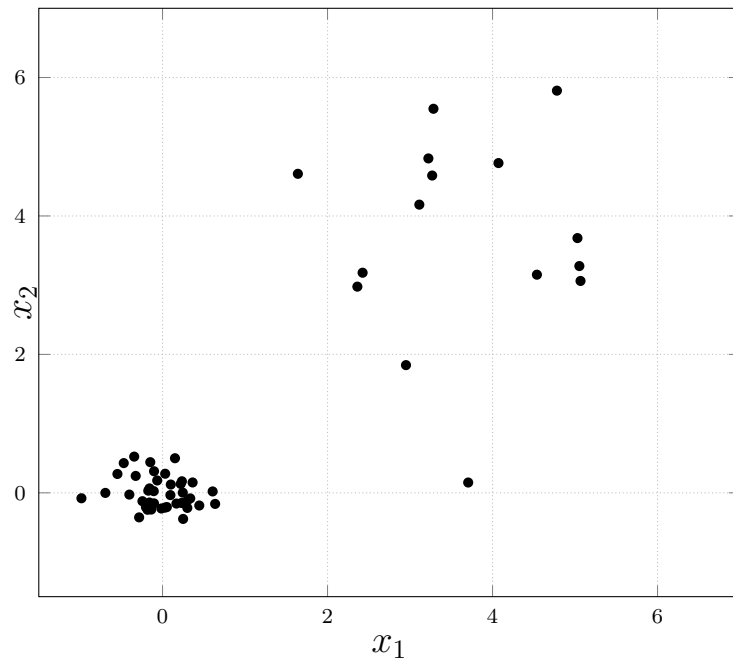
(c) Iteration 2



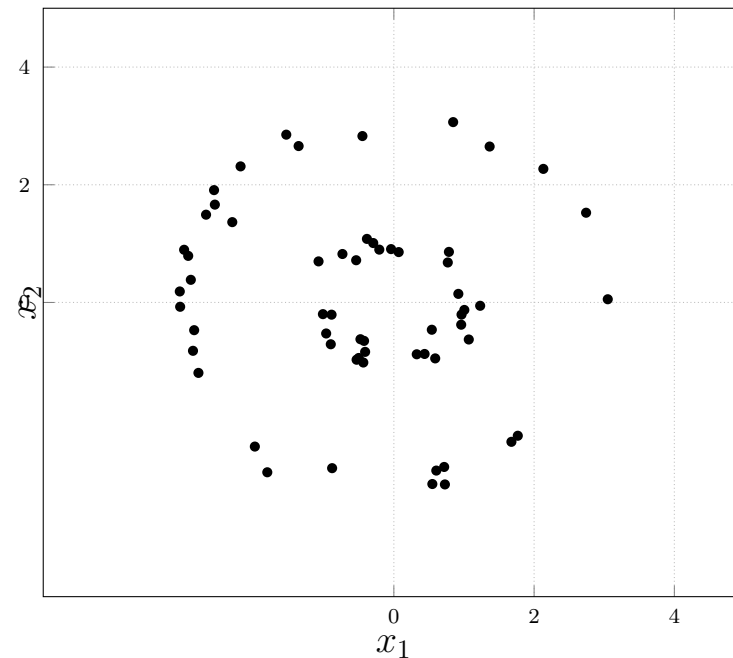
(d) Iteration 3 (converged)

Fig. 3: K -means algorithm on a data set with $m = 10$ and $k = 2$, starting from a poor initialisation. The crosses denote the representative vectors.

Potential Failures



(a) Clusters with different densities



(b) Concentric rings

Fig. 4: Limitations of K -means clustering.

Acknowledgement.: The ideas, exposition, and pedagogical flow in these notes are the author's own. The data sets used to illustrate the concepts were generated using Claude (Anthropic) based on specifications provided by the author.